

Users > moshika > Desktop > lab > 🐞 diagnose.pl

```
1  % ----- Knowledge base -----
2  % disease(Name, [list_of_key_symptoms], AdviceShort).
3
4  disease(flu,      [fever, cough, bodyache, sore_throat, fatigue],
5  |      'Rest, fluids, antipyretic (paracetamol), see physician if severe').
6  disease(common_cold, [sneezing, runny_nose, sore_throat, cough],
7  |      'Rest, fluids, decongestant if required').
8  disease(covid19, [fever, cough, loss_of_taste_smell, fatigue, breathlessness],
9  |      'Isolate, test (RT-PCR/antigen), seek medical help if breathless').
10 disease(malaria, [fever, chills, sweating, headache, malaise],
11 |      'Get malaria test (RDT/microscopy), antimalarial under doctor supervision').
12 disease(dengue, [fever, joint_pain, rash, bleeding, headache],
13 |      'Hydration, medical evaluation, platelet monitoring').
14 disease(typhoid, [fever, abdominal_pain, headache, constipation, diarrhea],
15 |      'Get Widal/ blood culture, antibiotics under doctor supervision').
16 disease(asthma, [wheezing, breathlessness, cough, chest_tightness],
17 |      'Inhaler (bronchodilator) as prescribed, avoid triggers').
18 disease(allergy, [sneezing, itchy_eyes, runny_nose, rash],
19 |      'Antihistamines, remove trigger, see allergist if needed').
20 disease(gastritis, [abdominal_pain, nausea, vomiting, indigestion],
21 |      'Avoid spicy/irritating foods, antacids, consult doctor').
22
23 % ----- Utility predicates -----
24
25 % intersect_count(List1, List2, Count)
26 % Count = number of unique elements common to List1 and List2
27 intersect_count(L1, L2, Count) :-
28 |     findall(X, (member(X, L1), member(X, L2)), Raw),
29 |     sort(Raw, Common),      % remove duplicates
30 |     length(Common, Count).
31
32 % make_pair_for_sort(Disease, Symptoms, NegScore-Disease-Score)
33 % We use NegScore so keysort gives descending Score order
34 make_pair_for_sort(Name, KBSymptoms, InputSymptoms, Key) :-
35 |     intersect_count(KBSymptoms, InputSymptoms, Score),
36 |     Neg is -Score,
37 |     Key = Neg-Name-Score.
38
39 % gather_matches(InputSymptoms, Pairs)
40 % Pairs = sorted list Neg-Name-Score (keysorted ascending -> highest Score first)
41 gather_matches(Symptoms, SortedPairs) :-
42 |     findall(Key, (disease(Name, KBSymptoms, _), make_pair_for_sort(Name, KBSymptoms, Symptoms, Key)), Pairs),
43 |     keysort(Pairs, SortedPairs).
```

```

44 |
45 | % ----- Output helpers -----
46 |
47 | print_match_list([]) :-
48 |     write('No matching diseases found. Consider consulting a physician. '), nl.
49 |
50 | print_match_list([Neg-Name-Score | Rest]) :-
51 |     Score > 0,
52 |     disease(Name, _, Advice),
53 |     RealScore is Score,
54 |     format('~nDiagnosis candidate: ~w (matching symptoms: ~w)~n Advice: ~w~n', [Name, RealScore, Advice]),
55 |     print_match_list(Rest).
56 |
57 | print_match_list([_--0 | _]) :- % remaining entries have score 0
58 |     write('No more matching candidates (remaining diseases share 0 symptom). '), nl.
59 |
60 | % ----- Interactive entry point -----
61 | % Enter symptoms as a Prolog list of atoms, e.g. [fever, cough, fatigue].
62 |
63 | start :-
64 |     write('Enter your symptoms as a Prolog list (e.g. [fever, cough, fatigue].) '), nl,
65 |     read(Symptoms),
66 |     ( is_list(Symptoms) ->
67 |         gather_matches(Symptoms, Sorted),
68 |         ( Sorted = [] ->
69 |             write('No diseases in knowledge base. '), nl
70 |             ; print_match_list(Sorted)
71 |         )
72 |     ; write('Please enter symptoms as a list, e.g. [fever, cough]. '), nl
73 |     ).
74 |
75 | % ----- Convenience query -----
76 | % diagnose(Symptoms, Results) gives back the list of Name-Score pairs (descending)
77 | diagnose(InputSymptoms, Results) :-
78 |     gather_matches(InputSymptoms, Sorted),
79 |     findall(Name-Score,
80 |         ( member(_Neg-Name-Score, Sorted), Score > 0 ),
81 |         Results).

```



```
[moshika@Moshikas-MacBook-Air ~ % cd /Users/moshika/Desktop/lab
[moshika@Moshikas-MacBook-Air lab % swipl diagnose.pl
Warning: /Users/moshika/Desktop/lab/diagnose.pl:50:
Warning: Singleton variables: [Neg]
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.
```

```
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
[?- start.
Enter your symptoms as a Prolog list (e.g. [fever, cough, fatigue].)
[: [fever, chills, headache, sweating].
```

```
Diagnosis candidate: malaria (matching symptoms: 4)
Advice: Get malaria test (RDT/microscopy), antimalarial under doctor supervision
```

```
Diagnosis candidate: dengue (matching symptoms: 2)
Advice: Hydration, medical evaluation, platelet monitoring
```

```
Diagnosis candidate: typhoid (matching symptoms: 2)
Advice: Get Widal/ blood culture, antibiotics under doctor supervision
```

```
Diagnosis candidate: covid19 (matching symptoms: 1)
Advice: Isolate, test (RT-PCR/antigen), seek medical help if breathless
```

```
Diagnosis candidate: flu (matching symptoms: 1)
Advice: Rest, fluids, antipyretic (paracetamol), see physician if severe
No more matching candidates (remaining diseases share 0 symptom).
true .
```